

The sacral splanchnic bypass: an anatomically specified and falsifiable refinement of the sympathetic hypothesis of hard flaccid syndrome

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¹HFS Patient • ²Large language model, Anthropic PBC

Abstract

Background. Hard flaccid syndrome (HFS) is a poorly characterised condition in which the flaccid penis is persistently semi-rigid, usually accompanied by sensory, urinary and pelvic-floor complaints. The most developed mechanistic account, proposed by Goldstein and Komisaruk, holds that HFS follows pathological activation of a somato-visceral “pelvic/pudendal-hypogastric” reflex, in which excess sympathetic efferent traffic reaches cavernosal smooth muscle through the hypogastric nerve and maintains it in contraction. That account names a specific efferent conduit. Whether that conduit is in fact the operative one has never been tested.

Hypothesis. We propose, and specify in falsifiable form, that a clinically meaningful proportion of the pathological sympathetic traffic in HFS reaches the erectile tissue through the *sacral splanchnic nerves*, which arise from the sacral portion of the sympathetic trunk and enter the inferior hypogastric plexus independently of the hypogastric nerves. We term this the *sacral splanchnic bypass*. Under a strong form the sacral route is sufficient alone; under a weak form it is a necessary co-contributor, so that interrupting either route in isolation is insufficient.

Basis. The dual sympathetic supply of the inferior hypogastric plexus is established human anatomy, although standard reference texts disagree about which route is quantitatively dominant. Rodent electrophysiology shows that stimulation of the pelvic nerve, the hypogastric nerve *or* the lumbosacral sympathetic chain each evokes activity in the penile nerve; that postganglionic sympathetic fibres reach the dorsal nerve of the penis directly from the chain; and that section of the hypogastric nerve alone does not measurably alter erectile responses, whereas section of the chain at L4–L5 does — although that last finding is confounded by a systemic haemodynamic effect and we do not treat it as decisive. Reported failure of superior hypogastric plexus blockade in HFS motivated this analysis, but those reports are unpublished, unverified and lack documentation of block adequacy; the hypothesis does not depend on them.

Predictions and tests. The hypothesis predicts a specific dissociation between anatomically targeted sympathetic blocks. We set out the interruption profile of each candidate block, show which comparisons discriminate the hypothesis from the standard model and which do not, and specify the two methodological requirements that all existing reports lack: objective confirmation of sympathetic block adequacy in a penile rather than a lower-limb territory, and an objective measure of resting cavernosal tone. No such validated tone measure currently exists; developing one is the rate-limiting step.

Conclusion. The sacral splanchnic bypass is a speculative but anatomically grounded and directly testable hypothesis. Nothing proposed here has been demonstrated in a patient with HFS, and nothing here should influence clinical management.

Keywords: hard flaccid syndrome; sacral splanchnic nerves; inferior hypogastric plexus; sympathetic nervous system; corpus cavernosum; genito-pelvic dysesthesia; sympathetic blockade.

Reader advisory

This is an exercise in hypothesis generation, not a report of new data and not a summary of established knowledge. No component of the proposed mechanism has been demonstrated in a person with hard flaccid syndrome. The interventions discussed — sympathetic nerve blocks in particular — are invasive procedures with real risks, are discussed here solely as research instruments, and are not recommended for clinical use in this condition. Readers should assume the hypothesis is more likely false than true; its value lies in being stated precisely enough to be refuted.

1. Introduction

Hard flaccid syndrome (HFS) is an acquired condition in which the penis remains in an unwanted, persistently semi-rigid flaccid state, commonly with penile pain, altered penile sensation, urinary symptoms, pelvic floor hypertonicity and marked psychological distress [1, 6, 4]. It is described predominantly in young men; in the largest symptom survey to

date the mean age was 27.4 years ($n = 143$) [5]. The condition is not represented in current diagnostic classifications, has no agreed diagnostic criteria, no validated objective test, and no controlled trial of any intervention. A 2025 PRISMA-registered systematic review identified only eight eligible studies, most of them case reports or congress abstracts, and rated the risk of bias as high [4].

The most developed mechanistic proposal is that of Goldstein and Komisaruk, who suggest that HFS results from pathological activation of a somato-visceral and/or viscero-visceral reflex that they term the *pelvic/pudendal-hypogastric reflex* [7, 8]. In this account, an afferent insult — arising at the penis, the pelvis or perineum, the cauda equina, the spinal cord or supraspinally — reflexively drives excess sympathetic efferent activity in the hypogastric nerve, which contracts cavernosal smooth muscle and produces the hard flaccid state. The proposal is coherent, is consistent with the reported worsening on standing and with the association between HFS and lumbosacral annular tears [8, 10], and is currently the field's principal organising idea.

It also makes a specific anatomical commitment. It names the hypogastric nerve as the efferent conduit. That commitment has never been tested, and it is separable from the rest of the hypothesis: one can accept that HFS is a sympathetically maintained contraction driven by a somato-autonomic reflex while doubting that the traffic travels the route assumed. The pelvic viscera receive sympathetic supply by more than one anatomical path, and the paths are separately interruptible. Which one carries the pathological signal is an empirical question with a practical consequence, because the two paths are targeted by different procedures at different vertebral levels.

This paper proposes and specifies an alternative efferent route. We call it the **sacral splanchnic bypass**. We state it formally, grade the evidence for and against each of its components, distinguish carefully between what is established anatomy, what is animal physiology, what is inference and what is unverified anecdote, and set out the experiments that would discriminate it from the standard account. We also identify the methodological obstacle that currently makes all such experiments uninterpretable, and argue that removing that obstacle is more urgent than testing any particular mechanism.

1.1 What this paper does not claim

We do not claim that the sacral splanchnic bypass is the mechanism of HFS. We do not claim it is more likely than the standard account. We do not claim that HFS is definitely a sympathetically maintained condition; that premise is inherited from the standard account and is itself unverified, since sympathetic outflow to the penis has never been measured in an affected person. We make no treatment recommendation of any kind.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated and no patients were involved; ethical approval was therefore not required.

2.1 Literature identification

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *hard flaccid syndrome*, *sacral splanchnic*, *inferior hypogastric plexus*, *superior hypogastric plexus block*, *lumbar sympathetic block*, *penile sympathetic innervation*, *cavernous nerve* and *dorsal nerve of the penis*, supplemented by backward citation chasing from the retrieved reviews. This was a purposive, non-systematic search intended to identify the anatomical and physiological literature bearing on a specific mechanistic question. It was not a systematic review, no PRISMA process was followed, no protocol was registered, and publication and language bias were not assessed. Readers should assume that relevant work was missed.

Where a primary source could not be retrieved in full, the claim is attributed to the secondary source in which it was encountered and flagged accordingly. Claims that we could not verify against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

2.2 Evidence grading

Because this argument mixes evidence of very unequal strength, every substantive claim below is labelled with the grade of evidence supporting it (Table 1). The grading refers to the strength of the evidence for the *stated proposition*, not to its relevance to HFS. No proposition in this paper carries grade A evidence *in HFS*, because no such evidence exists.

3. Background

3.1 The flaccid state is an actively maintained contraction

The normal flaccid penis is not a resting organ. Cavernosal and helicine arterial smooth muscle is held in tonic contraction, and erection is produced by relaxing it; noradrenaline acting at α -adrenoceptors is the principal contractile transmitter,

Table 1. Evidence grading scheme used throughout this paper.

Grade	Definition
[A]	Replicated human evidence: consistent findings across independent human studies, or uncontested descriptive human anatomy.
[B]	Single human study, human reference text, congress abstract, or human data with substantial methodological limitation.
[C]	Experimental animal evidence, with the species and preparation stated.
[D]	Mechanistic or anatomical inference: a conclusion deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, provenance uncertain, denominators unknown.

and the corpus cavernosum carries a dense adrenergic innervation [A] [13, 14, 15]. It follows that a pathologically firm flaccid penis is most economically read as an excess of the same process that maintains ordinary flaccidity. This inference is the load-bearing premise of every sympathetic account of HFS, including the one proposed here [D]. It is consistent with the reported clinical picture — a shrunken, contracted, non-compressible penis with a cold glans [7, 6] — but it has not been confirmed by direct measurement, and we return to this in Section 10.

3.2 The inferior hypogastric plexus has two sympathetic sources

The final autonomic relay to the human male pelvic organs is the inferior hypogastric plexus (IHP, or pelvic plexus), a fibro-nervous sheet lying posterolateral to the seminal vesicles, prostate and bladder base, from which the cavernous nerves arise and pass to the erectile tissue [A] [16, 14]. The IHP receives parasympathetic input from the pelvic splanchnic nerves (S2–S4) and sympathetic input by *two anatomically separate routes* [A] [16, 17]:

Route A. The prevertebral route. Preganglionic fibres arising from the intermediolateral column at approximately T10–L2 enter the sympathetic trunk, leave it as lumbar splanchnic nerves, traverse the intermesenteric and inferior mesenteric plexuses and the superior hypogastric plexus (SHP), and descend as the paired hypogastric nerves to the IHP.

Route B. The paravertebral route. Other preganglionic fibres of the same origin descend *within* the sympathetic trunk, pass without relay through the lumbar chain to the sacral sympathetic ganglia, and leave the trunk as the *sacral splanchnic nerves*, which pass medially and enter the IHP directly.

Both routes converge on the IHP, and postganglionic fibres from the IHP supply the pelvic viscera, genital ducts and external genitalia [A] [16, 17]. The functional distinction that matters here is that *route B never passes through the superior hypogastric plexus*.

3.3 Reference sources disagree about which route dominates

This is important enough to state plainly, because the standard account implicitly assumes route A is dominant and no primary quantitative human data settle the question.

One widely used reference text states that of the two sympathetic contributions to the IHP, “the largest contribution comes from the superior hypogastric plexus” [B] [16]. A second major anatomical reference states the opposite emphasis: that the preganglionic fibres travelling as lumbar splanchnics to the prevertebral ganglia and thence via the hypogastric nerve constitute *a minority*, with the remainder descending in the trunk to the sacral ganglia as sacral splanchnic nerves [B] [18]. The same source that reports the SHP as dominant also notes that classical anatomical studies provide few details of IHP morphology, that dissection is difficult, and that the relationships remain incompletely defined [16].

We could not identify a primary human dissection study that quantifies the relative fibre contribution of the two routes to the male IHP [DATA NEEDED]. This is a striking gap. The standard mechanistic account of HFS rests on a premise about relative route dominance that the anatomical literature does not agree on and has not measured.

3.4 What a superior hypogastric plexus block does and does not interrupt

The SHP is a retroperitoneal midline structure, usually anterior to the lower third of the L5 vertebral body and extending to the upper sacrum, from which the hypogastric nerves descend [A] [16]. Percutaneous SHP blockade was introduced for pelvic cancer pain and remains a standard interventional technique [B] [30]. Anatomically, a technically successful SHP block interrupts *route A only*. It is upstream of the IHP but downstream of nothing that reaches the IHP by the sacral splanchnic route.

Lumbar sympathetic blockade (LSB), by contrast, is applied to the paravertebral trunk itself, most commonly at the lower third of L2 or upper third of L3, where the lumbar sympathetic ganglia are densest [B] [31]. A conduction block at that point on the trunk interrupts fibres descending toward *both* the lumbar splanchnic origins of route A and the

sacral ganglia of route B. LSB is therefore a superset of SHP block with respect to pelvic sympathetic outflow — a fact that turns out to determine which clinical comparisons are informative and which are not (Section 7).

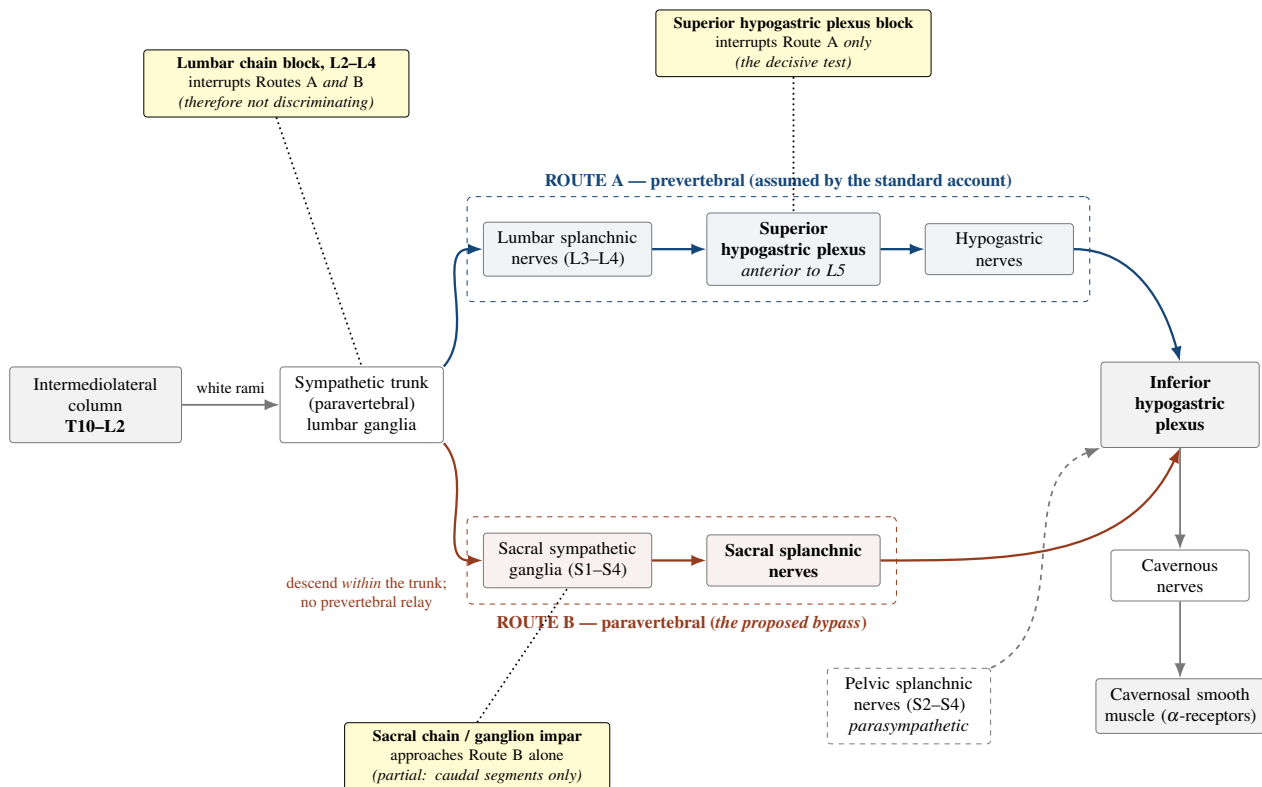


Figure 1. The two sympathetic routes from the thoracolumbar cord to the erectile tissue, and the level at which each candidate intervention interrupts them. Preganglionic fibres of common origin (T10–L2) diverge at the sympathetic trunk. Route A leaves the trunk as lumbar splanchnic nerves and reaches the inferior hypogastric plexus via the superior hypogastric plexus and the hypogastric nerves. Route B descends within the trunk to the sacral ganglia and enters the inferior hypogastric plexus directly as sacral splanchnic nerves, *without passing through the superior hypogastric plexus*. The standard account of hard flaccid syndrome implicitly assumes Route A; the hypothesis advanced here is that Route B carries a clinically meaningful share of the pathological traffic. Note that a lumbar chain block interrupts both routes and therefore cannot, by itself, distinguish them — which is why the superior hypogastric plexus block, and not the lumbar block, is the informative procedure. Parasympathetic input is shown dashed for orientation only. Diagram is schematic and not to anatomical scale.

4. The observation that motivates the hypothesis, and how much weight it can bear

4.1 The reports

A body of treatment-response observations circulates among patients and clinicians working with HFS, collated in the source article on which this paper builds [12]. Two elements are relevant here: superior hypogastric plexus blocks are reported to have been mostly unsuccessful, while lumbar sympathetic chain blocks at L2–L4 are reported to have resolved the hard flaccid state on a small number of occasions; a block at L1 is reported to have been unsuccessful; and in one patient an L2 block is reported to have reduced scrotal retraction without altering the hard flaccid state [E] [12].

4.2 Why these reports cannot support a conclusion

We state the objections in full, because the temptation to reason from them is exactly what this section is designed to resist.

1. **They are unpublished and unverified.** They are not peer reviewed, the sources cannot be independently checked, and they cannot be confirmed here.
2. **Block adequacy was never documented.** Not one reported block failure is recorded as a confirmed, technically adequate block. A block that did not reach its target is a failure of the procedure, not of the theory behind it. Failure to achieve adequate spread is common: in one series of lumbar sympathetic ganglion blocks, contrast was injected into psoas muscle in 21.3% of procedures and intravascularly in 12.5% [B] [VERIFY] [32].

3. **The outcome measure was patient impression.** There is no objective measure of penile resting tone in any of these reports.
4. **The denominators are unknown.** We do not know how many blocks of each type were performed, so “mostly unsuccessful” and “occasionally resolved” cannot be converted into rates.
5. **Selection and reporting bias.** Dramatic outcomes are more likely to be shared. Community reporting on HFS has itself been studied formally and is known to be a rich but non-representative data source [B] [2].
6. **Regression and placebo.** An invasive procedure performed at a moment of maximal symptom severity will sometimes be followed by improvement for reasons unrelated to the procedure.

4.3 Why the hypothesis is nonetheless worth stating

The hypothesis does not require the reports to be true. Its motivation is independent and stronger: *nobody has established which route carries sympathetic traffic to the human penis under any condition, normal or pathological*, and the anatomical literature does not agree (Section 3.3). The standard account of HFS names a conduit without evidence that it is the operative one. Naming the alternative, and specifying how the two could be told apart, is useful whether or not any particular anecdote is accurate. We use the reports strictly as a prompt, and no conclusion below rests on them.

5. The hypothesis

Formal statement

H1 (sacral splanchnic bypass). In hard flaccid syndrome, a clinically meaningful proportion of the sympathetic efferent traffic that maintains the pathological cavernosal contraction is conveyed to the inferior hypogastric plexus by the sacral splanchnic nerves arising from the sacral sympathetic trunk, rather than exclusively by the hypogastric nerves descending from the superior hypogastric plexus.

H1-strong. The sacral splanchnic route is *sufficient alone* to maintain the hard flaccid state. Prediction: a technically confirmed superior hypogastric plexus block does not abolish the state; a block that interrupts the sacral route does.

H1-weak. The sacral splanchnic route is a *necessary co-contributor*. Prediction: interruption of either route alone produces at most partial reduction in resting tone; interruption of both is required for abolition.

H0 (standard account). The hypogastric nerve is the operative conduit. Prediction: a technically confirmed superior hypogastric plexus block abolishes or substantially reduces the hard flaccid state.

Two auxiliary propositions are needed for H1-strong and are stated separately because each is a distinct liability:

- **A1 (saturation).** The relationship between total sympathetic drive and cavernosal tone is saturating rather than linear over the pathological range, so that a numerically minor route can produce a near-maximal effect when the tissue is already operating near the top of its contractile range. Without A1, a minority route cannot be sufficient. A1 is a plausible general property of smooth-muscle dose–response relations [D] but has not been characterised for human cavernosal tissue in vivo [DATA NEEDED].
- **A2 (target specificity).** The sacral splanchnic and hypogastric routes are not simply parallel copies of one another but supply partly distinct effector populations, so that dissociated outcomes are anatomically possible. Independent efferent populations to the dartos and to cavernosal tissue would be one instance [D].

6. Evidence bearing on the hypothesis

6.1 Human anatomy

The existence of the dual route is not in dispute [A] [16, 17]. Its relative magnitude is (Section 3.3) [B, contested].

A textbook account of the sympathetic supply of the penis, cited in a peer-reviewed review of neurogenic erectile dysfunction, states that although the preganglionic origin is thoracolumbar, *the sympathetic chain ganglia that innervate the penis are located in the sacral and caudal ganglia* [B] [VERIFY] [20, 19]. We were unable to retrieve the primary chapter and cite it as encountered in the secondary source; if accurate, it is close to a direct statement of the anatomical premise of H1, and confirming it against the primary text is a priority.

Sympathetic fibres reaching the penis are not confined to a single named nerve. Postganglionic sympathetic neurons supplying the penis are distributed across the sympathetic chain ganglia, the inferior mesenteric, hypogastric and pelvic ganglia, and possibly ganglia near the target organ, and sympathetic fibres are found in the pelvic, cavernous and pudendal nerves [B] [15]. Communicating branches between the dorsal, perineal and cavernous nerves of the penis have been demonstrated anatomically [B] [VERIFY] [28]. Communications between the inferior hypogastric plexus and the

puddendal nerves have been proposed as an explanation for incomplete relief of sympathetically mediated pelvic pain after plexus blockade and for partial preservation of function after surgical resection of the plexus [B] [VERIFY] [29].

The general lesson is that the pelvic autonomic supply is a redundant, anastomotic network rather than a set of private lines. A hypothesis that a single named nerve carries all of the pathological traffic is, on anatomical grounds alone, the less likely of the two on offer.

6.2 Animal electrophysiology

Three rodent findings bear directly on H1. All are grade C, and the species caveat in Section 6.5 applies to all of them.

1. **Three routes converge on the penile nerve.** In the anaesthetised rat, stimulation of the pelvic nerve, the hypogastric nerve *or* the sympathetic chain each elicited synaptic and axonal volleys in the penile nerve; the synaptic responses were abolished by nicotinic ganglionic blockade, locating the relay in peripheral ganglia [C] [21]. This is the clearest single demonstration that the chain provides a route to the penile efferent supply that is separate from the hypogastric nerve.
2. **The distal chain supplies penile nerve fibres directly.** Stimulation of the lumbar sympathetic chain at L4–L5 elicited an evoked discharge on the dorsal nerve of the penis, abolished by cutting the chain distal to the stimulation site; ganglionic blockade and serial neural sections demonstrated postganglionic sympathetic fibres within the dorsal nerve of the penis originating in the sympathetic chain [C] [22]. The route from the distal chain to the penis therefore exists and is functional in the rat, and it does not pass through the hypogastric nerve.
3. **The hypogastric nerve is dispensable; the distal chain is not.** In rats, erectile responses evoked by medial preoptic area stimulation were abolished by bilateral section of the cavernous or pelvic nerves or cauda equina, but *acute section of the hypogastric nerve did not significantly decrease the response*, whereas acute or chronic section of the paravertebral sympathetic chain at L4–L5, or chemical sympathectomy with 6-hydroxydopamine, significantly depressed it [C] [23]. Independently, bilateral transection of the hypogastric nerve alone in rats had no apparent pro- or anti-erectile effect across copulation, reflexive erection and non-contact erection [C] [VERIFY] [24].

6.2.1 An important interpretive caveat on finding 3

The depression of the erectile response after L4–L5 chain section was *reversed by bilateral ligation of the external iliac arteries* [23]. That reversal implies the effect was substantially haemodynamic — loss of vasoconstriction in the hindlimb vascular bed diverting flow away from the pelvis — rather than a direct loss of pro-erectile penile innervation. This must not be glossed over. What finding 3 establishes is asymmetric: the distal chain exerts substantial control over penile perfusion while the hypogastric nerve appears dispensable for this measure. It does not establish that the distal chain is the efferent conduit for cavernosal contractile tone. Read carefully, finding 3 is weaker support for H1 than it appears at first glance, and we do not treat it as decisive [D].

6.3 Route plasticity after pelvic nerve injury

H1 concerns HFS, which follows injury. Whether the operative route in an injured pelvis is the same as in an intact one is a separate question, and the answer in animals is that it need not be.

After chronic unilateral interruption of the pelvic nerve in rats, stimulation of the ipsilateral hypogastric nerve — which is ineffective in intact animals — began to elicit increases in penile pressure from three days onward [C] [25]. Comparable reorganisation of preganglionic connections has been described in cat bladder ganglia after parasymphatic denervation [C] [VERIFY] [26]. Uninjured lumbar preganglionic axons are reported to grow collaterals restoring central innervation to pelvic ganglion neurons after pelvic nerve transection [C] [VERIFY] [27].

The direction of this plasticity in the cited experiments is the opposite of what H1 requires — the hypogastric route gains function, rather than the sacral route. We therefore cite it only for the principle: **the efferent route to the pelvic viscera is not fixed, and can change within days of a pelvic nerve injury [C→D]**. If that principle holds in humans, then a route determined in a healthy pelvis is not automatically the route operating in an injured one, and any account of HFS that assumes a fixed conduit is assuming more than the evidence supports. This applies to H1 as much as to H0.

6.4 Clinical observations consistent with, but not specific to, H1

Two clinical facts fit H1 without discriminating it from alternatives.

First, the association between HFS and lumbosacral pathology. In a congress abstract, 76% (16/21) of men with HFS who underwent lumbar magnetic resonance imaging had a lumbosacral annular tear, most often at L5–S1 or L4–L5, and 38% had concomitant back pain or sciatica [B] [8, 4]; a case treated by lumbar endoscopic discectomy improved substantially [B] [7, 10]. This is consistent with a lumbosacral origin for the disturbance. It does not identify the efferent route, since a sacral radiculopathy could drive either route through a spinal reflex.

Second, the normality of penile Doppler studies. In the largest reported series ($n = 88$, congress abstract), all patients had normal peak systolic velocity, two had abnormal end-diastolic velocity, and none had elevated echogenicity on B-mode imaging [B] [11, 4]. (The proportion with abnormal end-diastolic velocity is reported as both 2.3% and 2.4% in the systematic review; with $n = 88$, $2/88 = 2.3\%$.) This argues against a primary arterial lesion and is compatible with a neurally maintained tone disorder, but it says nothing about which nerve carries the signal.

6.5 The species problem

Three of the four strongest pieces of evidence for H1 are from anaesthetised rats, and the extrapolation is not safe. The rat has a single, discrete major pelvic ganglion; the human inferior hypogastric plexus is a diffuse sheet whose fibres are difficult to trace and whose relative inputs have not been quantified [A] [16]. Segmental levels differ. Anaesthesia alters autonomic tone. Electrical stimulation of a nerve trunk demonstrates that a pathway *can* conduct, not that it *does* conduct under physiological or pathological conditions — a distinction that is central here, because H1 is a claim about which of two existing pathways carries traffic, not about which pathway exists. Every rodent finding above should be read as establishing anatomical possibility, not clinical actuality [D].

7. Predictions, and which comparisons actually discriminate

7.1 The interruption profile of each candidate intervention

The value of a comparison depends entirely on what each intervention interrupts. Table 2 sets this out. The critical and under-appreciated point is that **a lumbar chain block interrupts both routes**, so a dissociation between lumbar chain block and superior hypogastric plexus block is compatible with H1 but is not diagnostic of it: it shows only that something interrupted by the chain block and not by the plexus block matters. That something could be the sacral splanchnic route, but it could equally be lumbar visceral afferents, the sympathetic supply to non-cavernosal pelvic structures, or a systemic haemodynamic effect.

Table 2. Interruption profile of candidate interventions, and predicted outcome under each model.

Intervention	Route A	Route B	Also interrupts	Predicted under H0	Predicted under H1-strong
Superior hypogastric plexus block	Yes	No	Pelvic visceral afferents in hypogastric nerves	Tone falls	Tone unchanged
Lumbar chain block, L2–L4	Yes	Yes	Lower-limb sympathetic outflow; lumbar visceral afferents; systemic haemodynamics	Tone falls	Tone falls
Lumbar chain block, L1 only	Partial	Partial	—	Partial / none	Partial / none
Sacral chain or ganglion impar block	No	Partial (caudal)	Perineal afferents	Tone unchanged	Tone falls (partially)
Pudendal nerve block	No	No ^a	Somatic afferents and efferents	Tone unchanged	Tone unchanged

^a Except insofar as postganglionic sympathetic fibres travelling within the pudendal and dorsal penile nerves are blocked [15, 22]; the functional weight of that component in humans is unknown [DATA NEEDED].

7.2 The one comparison that discriminates

The single decisive test is **a technically confirmed superior hypogastric plexus block with an objective tone endpoint**. H0 predicts the tone falls. H1 predicts it does not. No auxiliary assumption is required, and the result cannot be explained away by route-B sparing under H0, because H0 asserts route B is not carrying the traffic.

A confirmatory but non-decisive second test is a block directed at the sacral sympathetic outflow alone. If tone falls when route B is interrupted and route A is intact, H1-strong gains substantial support. Anatomically this is the harder procedure and the interpretation is muddier, because a caudal sacral or ganglion impar approach reaches only part of route B.

7.3 Why the L1 failure is uninformative

The reported failure of an L1 block [E] has been treated as an anomaly. It is not. Preganglionic fibres destined for the pelvis arise across approximately T10–L2 and enter the trunk at their own segmental levels [A] [16, 17]. A single-level conduction block at L1 therefore spares fibres entering the trunk at L2, regardless of which downstream route they subsequently take. An incomplete result at L1 is predicted equally by H0 and H1 and discriminates nothing [D]. This is a small correction to the framing in the source article [12], which listed L1 failure as unexplained.

7.4 Two methodological requirements that all existing reports lack

7.4.1 Confirming block adequacy in the right territory

Sympathetic block adequacy is conventionally confirmed by a rise in skin temperature, typically at least 2 °C, in the somatic territory served [B] [31, 32]. For lumbar sympathetic block this is measured at the foot. **This is the wrong territory for the present question.** A foot temperature rise confirms interruption of sympathetic outflow to the lower limb; it does not confirm interruption of the visceral fibres descending toward the pelvis, and it is entirely uninformative for a superior hypogastric plexus block, which does not supply the limb at all.

Any study of sympathetic blockade in HFS should therefore document, as a minimum: (i) contrast spread at the intended target, recorded and independently read; (ii) a physiological index of sympatholysis in a *genital or perineal* territory — candidates include penile or perineal skin temperature, laser Doppler perfusion of the glans, perfusion index, or perineal sympathetic skin response; and (iii) explicit reporting of technical failures rather than their exclusion. Each of these candidate indices requires validation against a reference standard of pelvic sympatholysis before it can be relied upon [DATA NEEDED].

7.4.2 Measuring the thing that is supposed to change

This is the field's rate-limiting problem, and it is larger than any single hypothesis. **There is no validated objective measure of resting cavernosal tone in humans [DATA NEEDED].** Every reported treatment response in HFS, favourable or unfavourable, rests on patient impression of a sign the patient assesses by touch and appearance — an endpoint with no established reliability, no established minimal important difference, and obvious susceptibility to expectation effects, particularly after an invasive procedure.

Candidate approaches, none validated for this purpose, include: corpus cavernosum electromyography, whose reliability for individual diagnosis has long been contested [VERIFY]; shear-wave elastography of the corpora; penile durometry or applanation tonometry; graded-compression plethysmography to estimate flaccid compliance; and direct measurement of flaccid intracavernosal pressure, which is invasive and unlikely to be acceptable outside a procedure already indicated. Establishing a measure with known test–retest reliability in healthy controls, known responsiveness to a manipulation of known direction (for example, a standardised cold pressor test, which should raise sympathetic outflow), and known variance — from which a sample size could finally be calculated — would do more for this field than any mechanistic study. At present, no meaningful power calculation for any HFS intervention trial is possible [DATA NEEDED].

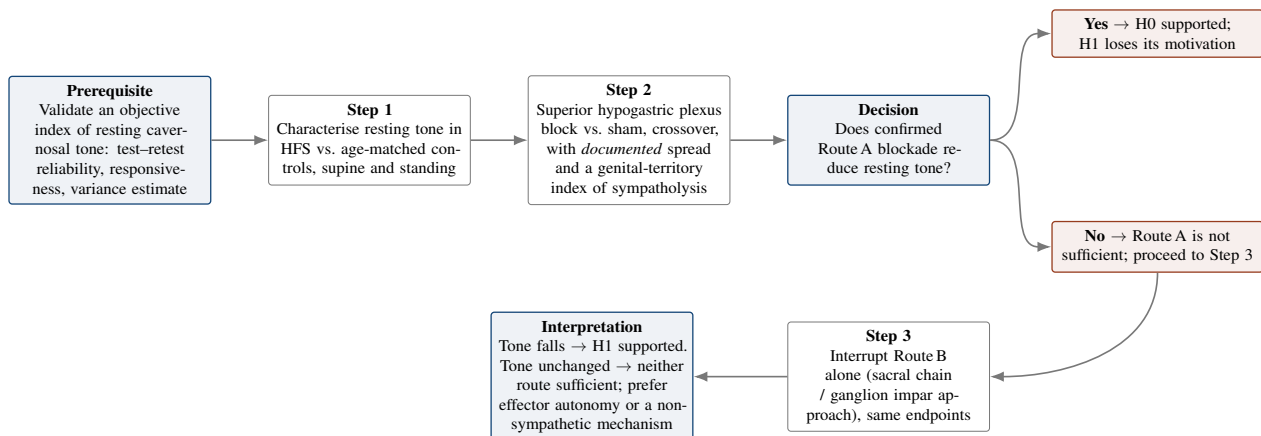


Figure 2. Proposed discriminating sequence. The prerequisite step is not optional. Without a validated objective tone endpoint, every downstream result is uninterpretable, which is the reason the existing block reports cannot settle the question they appear to address. All blocks are research procedures in this scheme, would require formal ethical approval, and are not proposed as treatment.

8. Alternative explanations

A hypothesis is only as good as its competitors, and several are at least as plausible as H1.

1. **Technical failure.** The most parsimonious explanation of reported superior hypogastric plexus block failure is that the blocks were inadequate. Given documented rates of misplacement in analogous procedures [32], this should be the default assumption until adequacy is documented. If it is correct, H1 loses its motivating observation entirely, though not its independent anatomical rationale.
2. **Effector autonomy.** The cavernosal smooth muscle may have become hypercontractile independently of ongoing

neural drive — through calcium sensitisation, denervation supersensitivity, co-transmitter dominance, or altered pacemaker activity in interstitial cells [12]. On this view no sympathetic block of any kind would abolish the state, and the route question would be moot. This class of explanation accounts for the same observation as H1 with a different lesion, and the two are distinguished by whether *any* confirmed sympatholysis reduces tone.

3. **Circulating catecholamines.** No regional block removes circulating adrenaline. A supersensitive or otherwise altered effector would continue to respond to humoral stimulation, producing apparent block failure under any anatomical model [D].
4. **Systemic and non-specific effects of lumbar blockade.** Lumbar sympathetic block at L2–L4 produces substantial lower-limb vasodilatation and hypotension [B] [31]. Any apparent benefit could reflect systemic haemodynamic change, spread to somatic structures, or the effect of a large-volume injection, rather than pelvic sympatholysis — exactly the confound that made the rodent chain-section result difficult to interpret [23].
5. **The premise itself.** HFS may not be a sympathetically maintained cavernosal contraction. It has never been shown that sympathetic outflow to the penis is elevated in an affected person, nor that cavernosal tone is objectively raised. If either premise is false, the entire class of hypotheses to which H1 belongs collapses. We regard this as the most serious unexamined risk in the field.
6. **Heterogeneity.** Only 58.0% of surveyed patients reported onset after a specific incident [5]. If the remaining 42.0% represent a different condition with a shared presentation, pooling them will obscure any true mechanism. Comparable heterogeneity may exist by symptom duration.

9. Implications

We deliberately draw no clinical implications. Two research implications follow.

First, reporting standards. Any future report of a sympathetic block in HFS — including negative and anecdotal ones — should state the anatomical target, the vertebral level, the volume and agent, the imaging confirmation of spread, the index and territory used to confirm sympatholysis, and the outcome measure. Reports lacking these cannot contribute to the question and, as the present analysis shows, actively mislead by appearing to.

Second, stratification. Studies should record and report symptom duration and whether onset followed an identifiable incident, and analyse accordingly. If route dominance changes after injury, as animal work suggests it can [25, 26], then pooling patients across durations would obscure the very effect under investigation.

10. Limitations

The limitations of this paper are severe and we state them without mitigation.

1. **No direct evidence in HFS.** Not one component of H1 has been demonstrated in a person with hard flaccid syndrome. Sympathetic outflow to the penis has never been measured in an affected person by any route.
2. **The motivating observation is grade E.** The block-response reports are unverified, undenominated, unblinded, and unaccompanied by any documentation of technical adequacy.
3. **Species extrapolation.** The strongest mechanistic support is rodent, from anaesthetised acute preparations using supramaximal electrical stimulation, in an animal whose pelvic ganglion architecture differs materially from the human.
4. **One key rodent finding is confounded.** The apparent importance of the L4–L5 chain for erectile responses was reversed by external iliac ligation, indicating a substantially haemodynamic mechanism [23].
5. **The anatomical premise is contested and unquantified.** Reference texts disagree on which route dominates, and we identified no primary human study quantifying the contributions.
6. **H1-strong requires an unverified auxiliary assumption.** Saturating dose–response behaviour (A1) has not been characterised for human cavernosal tissue in vivo.
7. **No endpoint exists.** Without a validated measure of resting cavernosal tone, the discriminating experiment cannot presently be performed, and no sample size can be calculated.
8. **Non-systematic search.** The literature base was assembled purposively; relevant work was probably missed, and no formal risk-of-bias assessment was applied to the sources.
9. **Several citations are secondary or partially verified,** and are marked [VERIFY] in the text. Readers should confirm them against the primary sources before relying on them.

10. **Confirmation risk.** This paper was written to develop a specific conjecture. That framing predisposes to selecting supportive evidence. We have tried to counter it by grading every claim, by stating the confound in the strongest supporting rodent study, and by giving the alternatives their own section, but the reader should discount accordingly.

11. Refutation criteria

We state the conditions under which the hypothesis should be abandoned.

- **Refuted (H1-strong).** A technically confirmed superior hypogastric plexus block, with documented spread and documented genital-territory sympatholysis, reliably abolishes or substantially reduces objectively measured resting cavernosal tone in HFS.
- **Refuted (H1 as a whole).** Confirmed interruption of the sacral sympathetic outflow produces no change in objectively measured tone in patients in whom sympathetic blockade at another level does.
- **Substantially weakened.** Primary human dissection studies establish that the sacral splanchnic contribution to the male inferior hypogastric plexus is negligible; or human tracing studies show that the postganglionic neurons supplying the corpora lie predominantly on the hypogastric route.
- **Rendered moot.** Direct measurement shows that sympathetic outflow to the penis is normal or reduced in symptomatic patients, or that resting cavernosal tone is not objectively elevated. Either finding would refute the class of hypotheses to which H1 belongs, not merely H1.

12. Conclusion

The prevailing account of hard flaccid syndrome names the hypogastric nerve as the efferent conduit for a pathological sympathetic reflex. That naming is an assumption, not a finding. The inferior hypogastric plexus receives sympathetic input by two anatomically separate routes; reference texts disagree about which dominates; no primary human study has quantified them; and rodent electrophysiology shows that the paravertebral route reaches the penile efferent supply independently of the hypogastric nerve, while section of the hypogastric nerve alone leaves erectile responses measurably unchanged.

The sacral splanchnic bypass hypothesis states that the paravertebral route carries a clinically meaningful share of the pathological traffic in HFS. It is speculative, it rests in part on an auxiliary assumption about dose–response saturation that has not been tested, and its motivating clinical observation is anecdotal and unverified. It has one virtue that recommends it: it is refutable by a single well-conducted experiment, and that experiment — a technically confirmed superior hypogastric plexus block with an objective tone endpoint — would be equally informative whichever way it came out, because it tests the standard account as directly as it tests this one.

That experiment cannot currently be performed, because no validated objective measure of resting cavernosal tone exists. Developing one is, in our view, a higher priority for this field than adjudicating between mechanisms. Until it exists, hard flaccid syndrome will continue to generate hypotheses faster than it can dispose of them, and this paper will have added one more.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. We list it as requested by the corresponding author and in the interest of transparency, but readers and editors should note that this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet the criteria for authorship — principally because they cannot take responsibility for the accuracy and integrity of the work or manage conflicts of interest — and that their use should instead be disclosed in the methods or acknowledgements. **Before submission to any peer-reviewed venue, this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** The human author should independently verify every citation, and in particular those marked [VERIFY], against the primary sources before submission.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

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Competing interests. The authors declare no competing financial interests. The first author maintains a public website on hard flaccid syndrome and is the author of the source article [12] from which the conjecture examined here is drawn; this constitutes a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation and does not constitute medical advice or establish a clinician–patient relationship. Nothing in it should be used to select, request or avoid any treatment. Sympathetic nerve blocks and the other interventions named are invasive procedures carrying real risk, appropriate only under qualified medical supervision and, in this

condition, only within an approved research protocol.

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